

StudyFlowAI

AI Prompts Used During Development

These are prompts from my actual sessions with Arena.ai Agent Mode while building this project (lightly cleaned for spelling; error messages are verbatim). My protocol throughout the build: one step at a time, paste real errors exactly as they appear, say “explain this part” whenever I didn’t understand something before moving on, and verify everything by running it myself.

Planning the project

- *“Here is my project requirements PDF (25 pages). Read it fully and tell me exactly what I need to build. Then ask me questions about my stack, deployment and experience before we start any code.”*
- *“Does the PDF force any particular tech stack? I know MERN — is MERN difficult for this project, or should I use something else?”*
- *“Let’s go with MERN then. Guide me step by step with code, from project setup to the end — but don’t dump the full project on me at once, I want to build and understand each part myself.”*
- *“Given the deadline, what should my 3-day plan look like? What are the riskiest parts I should build first?”*

Backend & database

- *“Set up the Express + MongoDB backend: env config that fails fast, a central error handler, JWT auth with bcrypt, and a health route. Explain why each file exists.”*
- *“Add Spaces and Projects CRUD. Ownership must be checked inside the database query itself and return 404 — never 403, so strangers can’t even confirm a project exists.”*
- *“Before you give me the next step: run the server and test each endpoint with curl, and only give me code that you verified works.”*

AI features (tutor, quiz, retrieval)

- *“Add a Gemini layer where every AI call is saved to an aicalls collection — model, latency, tokens, success/failure. Use batch embeddings for document chunks and a separate query embedding for questions.”*
- *“For the tutor: if the best similarity score is below 0.3, refuse to answer WITHOUT calling the model at all. Otherwise answer only from the retrieved chunks with [1] [2] citations.”*
- *“For the quiz: pick the 3 weakest concepts from mastery data, and generate MCQ + open-ended questions in ONE call to save quota. The model should return the correct option’s TEXT, not an index number — my code will derive the index.”*

Debugging — real errors, pasted as they came

- *“My upload is failing with this error: “models/gemini-2.0-flash is no longer available”. Fix it in a way that the next time a model is deprecated, I only change an environment variable.”*
- *“Tutor crashes with: “Cannot read properties of undefined (reading ‘values’)”. Explain the root cause first — don’t just patch over it.”*
- *“Now every Start-quiz click fails with “no valid questions after normalization” and the logs show correctIndex: -1 on the dropped items. Why is this happening?”*
- *“So models are bad at index positions. Change it so the model returns the answer text and my code derives the index instead.”*
- *“I am getting an internal server error in the quiz after changing my API key. Here are my Render logs [pasted full log tail]. What should I do?”*

Deployment & day-to-day practical questions

- *"If I change the Gemini key in my local project, do I also need to change it on Render? What about Vercel?"*
- *"Can we create many API keys in the free tier of Gemini, or does the quota stay the same?"*
- *"I have not deployed the frontend on Vercel yet — walk me through it step by step."*

Review before submission

- *"Did we build everything the requirements PDF asked for? Did we add any innovation, or only the baseline? Check my project against the PRD properly, point by point."*
- *"What exactly should I click and show in the demo video so every required feature is visible?"*

What I did not delegate: running the server, testing endpoints myself, reading the database, running the test suite on every change, deployments, and the final read of every file I submitted. AI drafted and suggested; I verified, decided and owned the result.